# Lab-2

# General Assembly Language Programming

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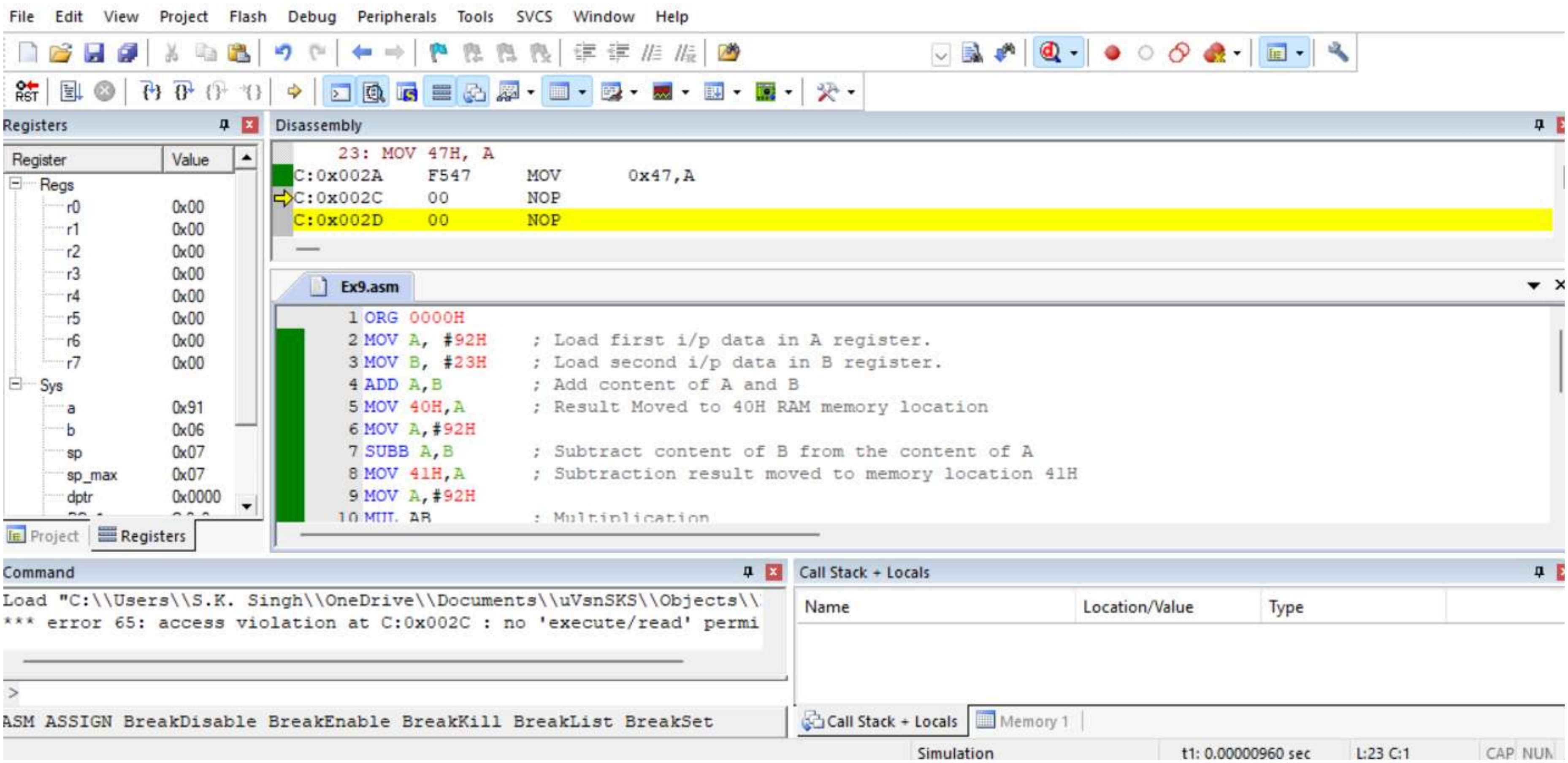
**Example:**

WAP for basic arithmetic operations (Addition, Subtraction, Multiplication, Division, Increment and Decrement) on the 8051 microcontroller on two numbers given as: 92H and 23H. Store the results in memory locations starting from 40H to 47H.

**Program:**

```
ORG 0000H
MOV A, #92H     ; Load first i/p data in A register.
MOV B, #23H     ; Load second i/p data in B register.
ADD A,B         ; Add content of A and B
MOV 40H,A       ; Result Moved to 40H RAM memory location
MOV A,#92H
SUBB A,B        ; Subtract content of B from the content of A
MOV 41H,A       ; Subtraction result moved to memory location 41H
MOV A,#92H
MUL AB          ; Multiplication
MOV 42H,A       ; Lower byte of result moved to memory location 42H
MOV 43H,B       ; Higher byte of result moved to memory location 43H
MOV A, #92H
MOV B, #23H
DIV AB
MOV 44H,A        ; Quotient moved to 44H memory location
MOV 45H,B        ; Remainder moved to 45H memory location
MOV A, #92H
INC A
MOV 46H, A
MOV A,#92H
DEC A
MOV 47H, A
END
```

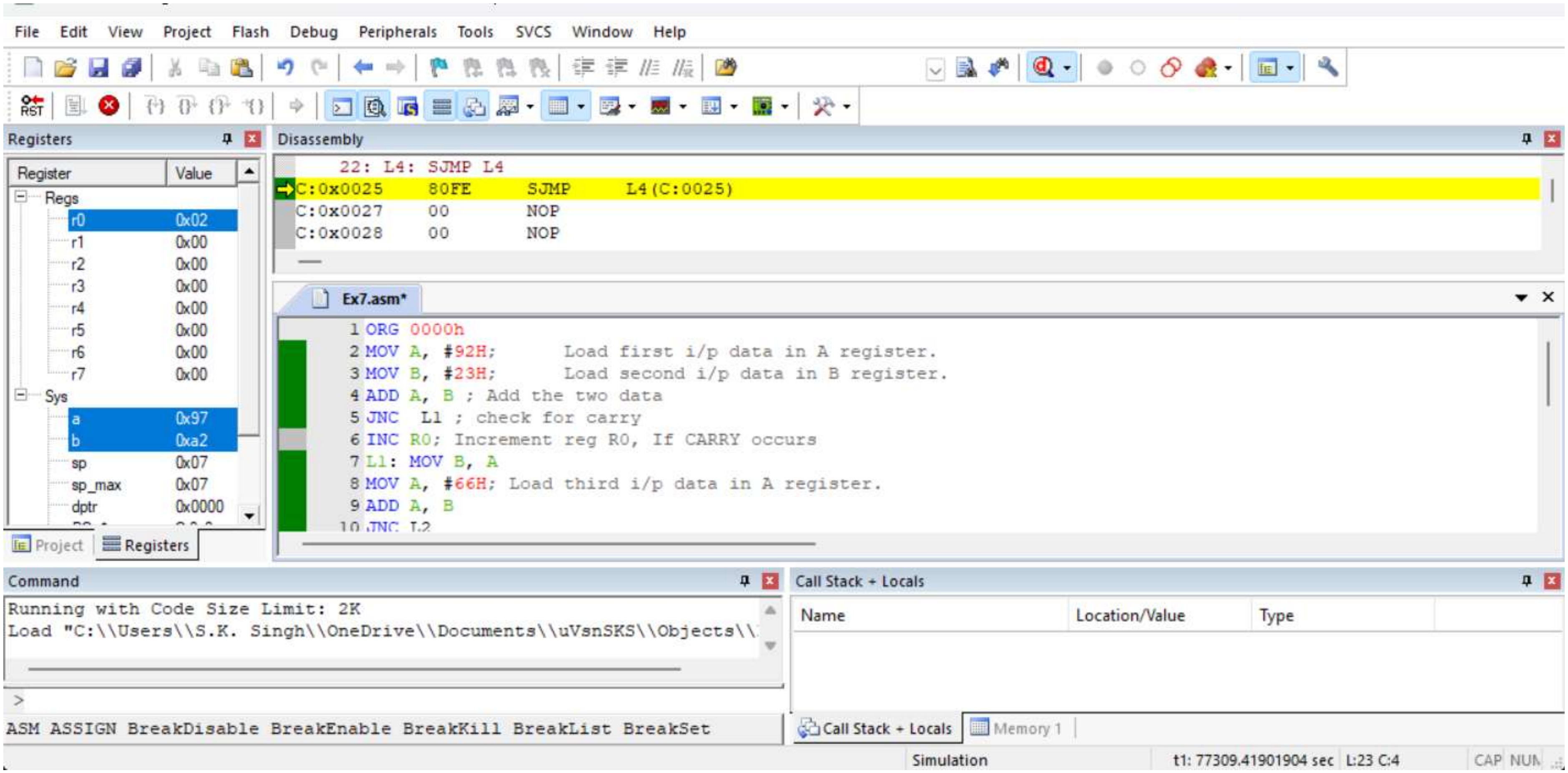

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Regs
r0 0x00
r1 0x00
r2 0x00
r3 0x00
r4 0x00
r5 0x00
r6 0x00
r7 0x00
Sys
a 0x91
b 0x06
sp 0x07
sp_max 0x07
dptr 0x0000
Project Registers
Disassembly
23: MOV 47H, A
C:0x002A F547 MOV 0x47,A
C:0x002C 00 NOP
C:0x002D 00 NOP
Ex9.asm
1 ORG 0000H
2 MOV A, #92H ; Load first i/p data in A register.
3 MOV B, #23H ; Load second i/p data in B register.
4 ADD A,B ; Add content of A and B
5 MOV 40H,A ; Result Moved to 40H RAM memory location
6 MOV A,#92H
7 SUBB A,B ; Subtract content of B from the content of A
8 MOV 41H,A ; Subtraction result moved to memory location 41H
9 MOV A,#92H
10 MUL AB ; Multiplication
Command
Load "C:\\Users\\S.K. Singh\\OneDrive\\Documents\\uVsnSKS\\Objects\\
*** error 65: access violation at C:0x002C : no 'execute/read' permi
ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet
Call Stack + Locals
Name Location/Value Type
Call Stack + Locals Memory 1
Simulation t1: 0.00000960 sec L:23 C:1 CAP NUM

**Example:**

Add five hexadecimal numbers: 92H, 23H, 66H, 87H, F5H. Use A (Accumulator) and B register for addition. Each time an addition produces a carry, increment R0. Final sum is left in A and B. R0 = total number of carries.

**Program:**

```
ORG 0000H
MOV A, #92H      ; Load 1st data in A register.
MOV B, #23H      ; Load 2nd data in B register.
ADD A, B         ; Add the two data
JNC  L1          ; Check for carry
INC R0           ; Increment register R0, If CARRY occurs
L1: MOV B, A     ; Move contents of A into B register.
MOV A, #66H      ; Load 3rd data in A register.
ADD A, B
JNC L2
INC R0
L2: MOV B,A
MOV A, #87H      ; Load 4th data in A register.
ADD A,B
JNC L3
INC R0
L3: MOV B,A
MOV A, #0F5H     ; Load 5th data in A register.
ADD A, B
JNC L4
INC R0
L4: SJMP L4
END
```


File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Regs
r0 0x02
r1 0x00
r2 0x00
r3 0x00
r4 0x00
r5 0x00
r6 0x00
r7 0x00
Sys
a 0x97
b 0xa2
sp 0x07
sp_max 0x07
dptr 0x0000
Project Registers
Disassembly
22: L4: SJMP L4
C:0x0025 80FE SJMP L4(C:0025)
C:0x0027 00 NOP
C:0x0028 00 NOP
Ex7.asm*
1 ORG 0000h
2 MOV A, #92H; Load first i/p data in A register.
3 MOV B, #23H; Load second i/p data in B register.
4 ADD A, B ; Add the two data
5 JNC L1 ; check for carry
6 INC R0; Increment reg R0, If CARRY occurs
7 L1: MOV B, A
8 MOV A, #66H; Load third i/p data in A register.
9 ADD A, B
10 JNC L2
Command
Running with Code Size Limit: 2K
Load "C:\\Users\\S.K. Singh\\OneDrive\\Documents\\uVsnSKS\\Objects\\
ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet
Call Stack + Locals
Name Location/Value Type
Call Stack + Locals Memory 1
Simulation t1: 77309.41901904 sec L:23 C:4 CAP NUM

**Example:**

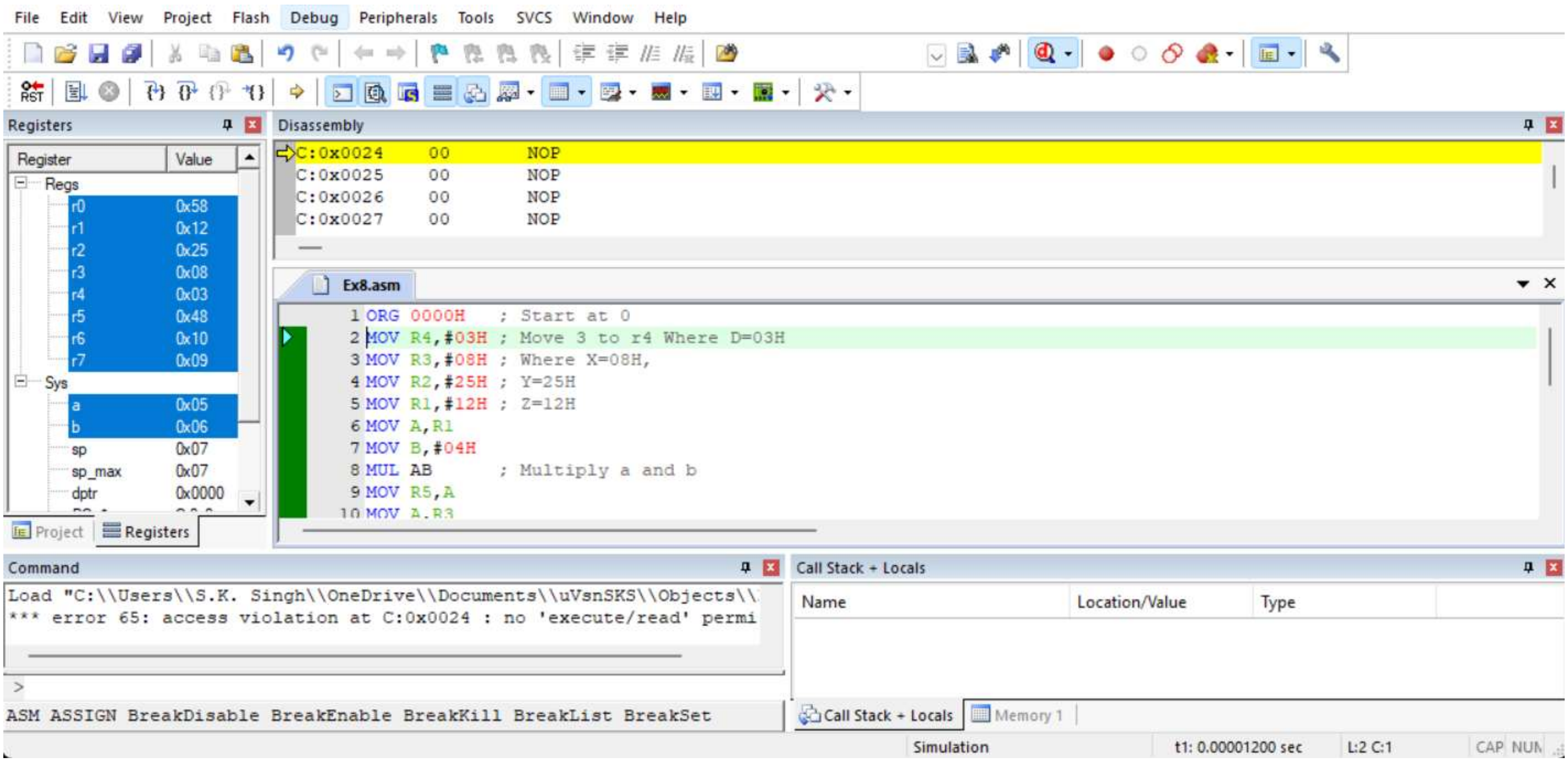
Write an 8051 ASM program to solve the following mathematical equation:

$$W = (4Z - Y + 2X)/3D$$

where, $D = 03H$, $X = 08H$, $Y = 25H$ and $Z = 12H$.

**Program:**

```
ORG 0000H    ; Start at 0
MOV R4,#03H ; Move 3 to r4 Where D=03H
MOV R3,#08H ; Where X=08H,
MOV R2,#25H ; Y=25H
MOV R1,#12H ; Z=12H
MOV A,R1
MOV B,#04H
MUL AB       ; Multiply a and b
MOV R5,A
MOV A,R3
MOV B,#02H
MUL AB
MOV R6,A
MOV A,R4
MOV B,#03H
MUL AB
MOV R7,A
MOV A,R5
ADD A,R6
MOV R0,A
MOV A,R0
SUBB A,R2
MOV B,R7
DIV AB
MOV 22H,A
END
```

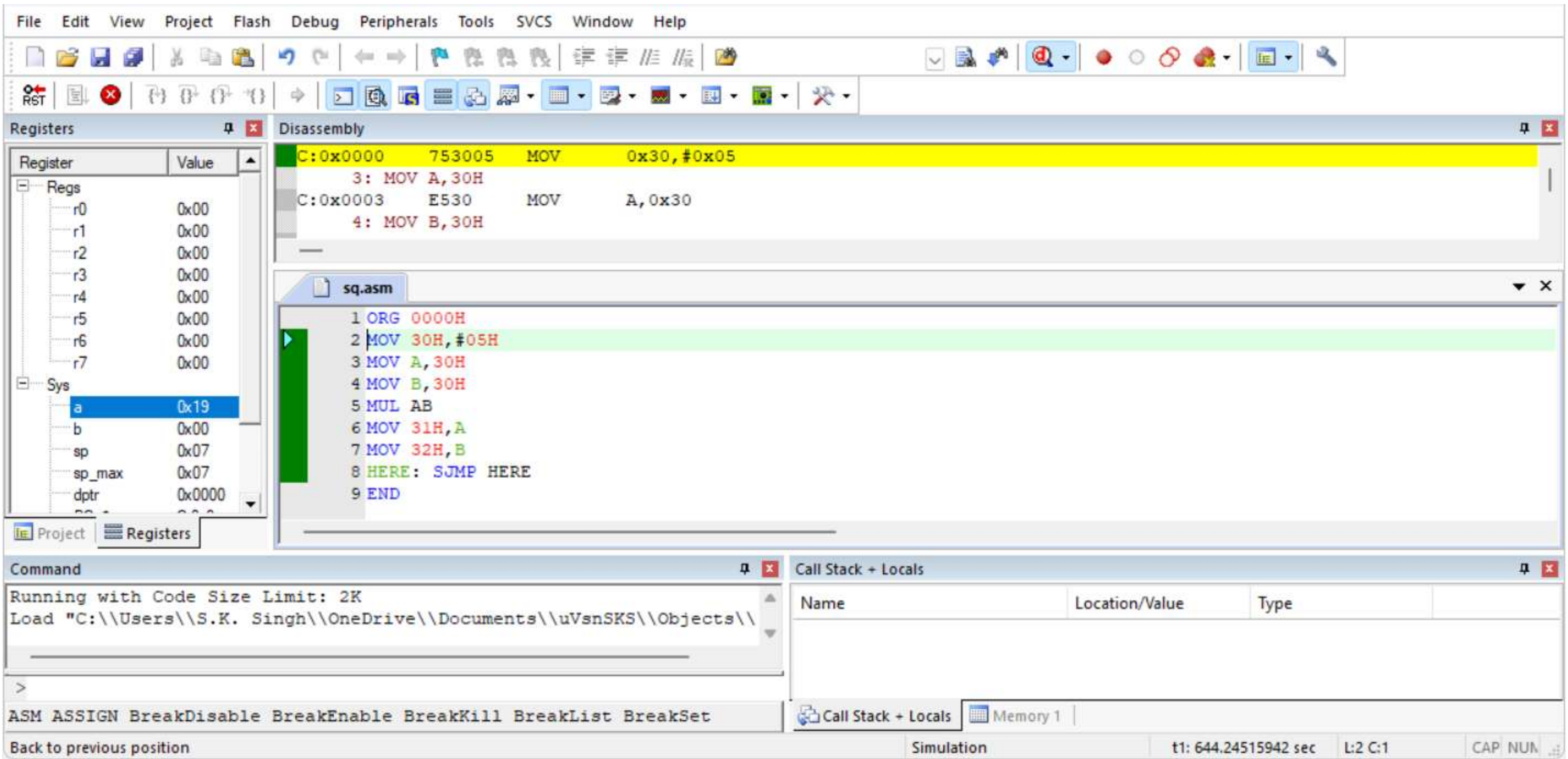

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register
Value
Regs
r0 0x58
r1 0x12
r2 0x25
r3 0x08
r4 0x03
r5 0x48
r6 0x10
r7 0x09
Sys
a 0x05
b 0x06
sp 0x07
sp_max 0x07
dptr 0x0000
Project
Registers
Disassembly
C:0x0024 00 NOP
C:0x0025 00 NOP
C:0x0026 00 NOP
C:0x0027 00 NOP
Ex8.asm
1 ORG 0000H ; Start at 0
2 MOV R4,#03H ; Move 3 to r4 Where D=03H
3 MOV R3,#08H ; Where X=08H,
4 MOV R2,#25H ; Y=25H
5 MOV R1,#12H ; Z=12H
6 MOV A,R1
7 MOV B,#04H
8 MUL AB ; Multiply a and b
9 MOV R5,A
10 MOV A,R3
Command
Load "C:\\Users\\S.K. Singh\\OneDrive\\Documents\\uVsnSKS\\Objects\\
*** error 65: access violation at C:0x0024 : no 'execute/read' permi
>
ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet
Call Stack + Locals
Name
Location/Value
Type
Call Stack + Locals
Memory 1
Simulation
t1: 0.00001200 sec
L:2 C:1
CAP NUM

**Example:**

Write an ALP to find square of a number.

**Program:**

```
ORG 0000H
MOV 30H,#05H
MOV A,30H
MOV B,30H
MUL AB
MOV 31H,A
MOV 32H,B
HERE: SJMP HERE
END
```

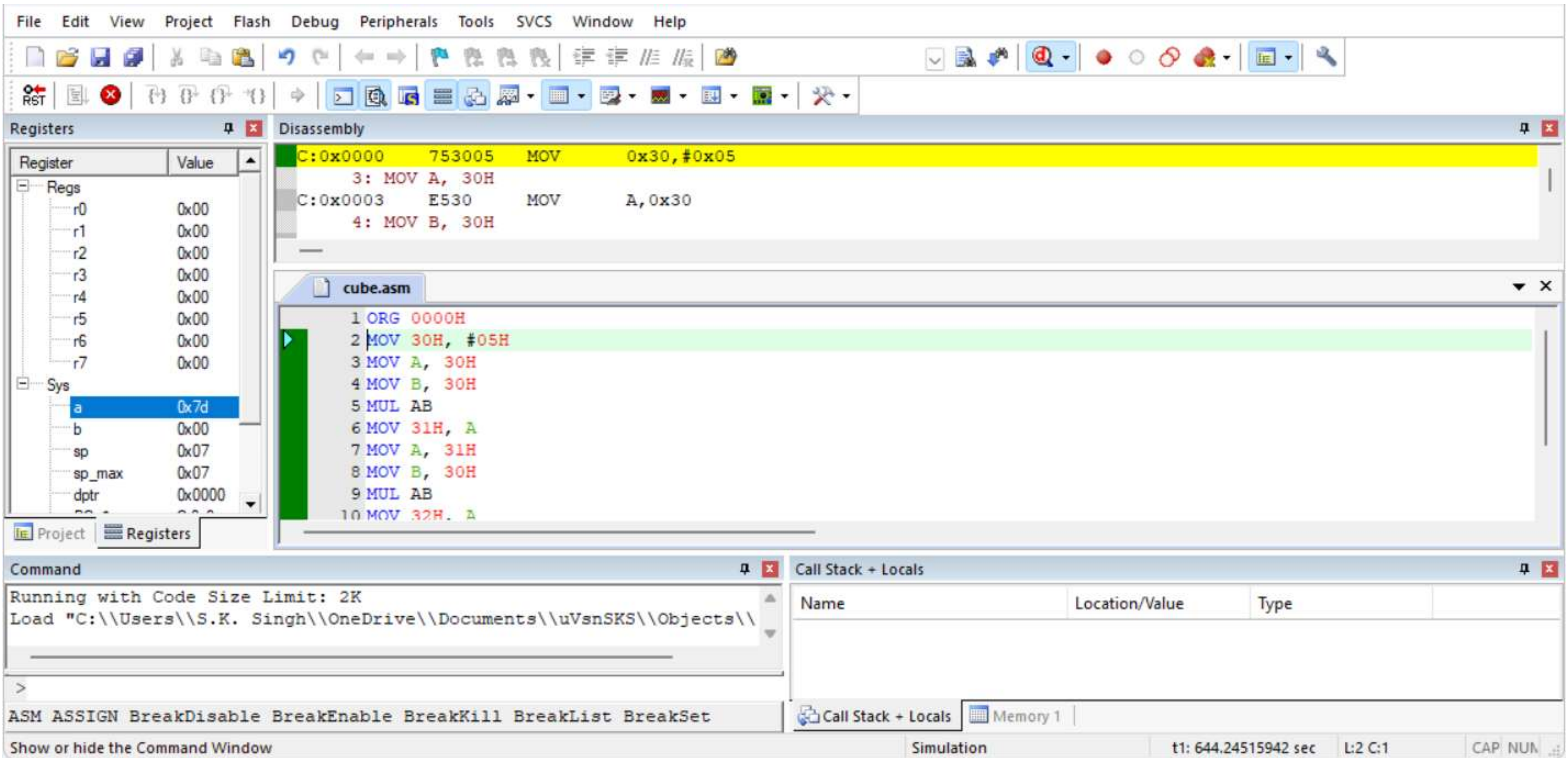

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Disassembly
C:0x0000 753005 MOV 0x30,#0x05
3: MOV A,30H
C:0x0003 E530 MOV A,0x30
4: MOV B,30H
Register Value
Regs
r0 0x00
r1 0x00
r2 0x00
r3 0x00
r4 0x00
r5 0x00
r6 0x00
r7 0x00
Sys
a 0x19
b 0x00
sp 0x07
sp_max 0x07
dptr 0x0000
sq.asm
1 ORG 0000H
2 MOV 30H,#05H
3 MOV A,30H
4 MOV B,30H
5 MUL AB
6 MOV 31H,A
7 MOV 32H,B
8 HERE: SJMP HERE
9 END
Project Registers
Command
Running with Code Size Limit: 2K
Load "C:\\Users\\S.K. Singh\\OneDrive\\Documents\\uVsnSKS\\Objects\\
ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet
Call Stack + Locals
Name Location/Value Type
Call Stack + Locals Memory 1
Back to previous position
Simulation
t1: 644.24515942 sec
L:2 C:1
CAP NUM

**Example:**

Write an ALP to find cube of a number.

**Program:**

```
ORG 0000H
MOV 30H,
#05H
MOV A, 30H
MOV B, 30H
MUL AB
MOV 31H, A
MOV A, 31H
MOV B, 30H
MUL AB
MOV 32H, A
MOV 33H, B
HERE: SJMP
HERE
END
```


File Edit View Project Flash Debug Peripherals Tools SVCS Window Help
Registers
Register Value
Regs
r0 0x00
r1 0x00
r2 0x00
r3 0x00
r4 0x00
r5 0x00
r6 0x00
r7 0x00
Sys
a 0x7d
b 0x00
sp 0x07
sp_max 0x07
dptr 0x0000
Project Registers
Disassembly
C:0x0000 753005 MOV 0x30,#0x05
3: MOV A, 30H
C:0x0003 E530 MOV A,0x30
4: MOV B, 30H
cube.asm
1 ORG 0000H
2 MOV 30H, #05H
3 MOV A, 30H
4 MOV B, 30H
5 MUL AB
6 MOV 31H, A
7 MOV A, 31H
8 MOV B, 30H
9 MUL AB
Command
Running with Code Size Limit: 2K
Load "C:\\Users\\S.K. Singh\\OneDrive\\Documents\\uVsnSKS\\Objects\\
ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet
Call Stack + Locals
Name Location/Value Type
Call Stack + Locals Memory 1
Show or hide the Command Window
Simulation
t1: 644.24515942 sec
L:2 C:1
CAP NUM